

Round 1

WARM-UP EXERCISE: **PAIR UP**



```
int arr[3] = {2, 4, 6};  
printf("%d\n", arr[2]);
```

```
int arr[3] = {3, 6, 9};  
int *p = arr;  
printf("%d\n", *(p + 1));
```

```
void change(int x) {  
    x = 100;  
}
```

```
int main() {  
    int a = 10;  
    change(a);  
    printf("%d", a);  
    return 0;  
}
```

```
void modify(int a[])  
{  
    a[0] = 10;  
}
```

```
int main()  
{  
    int arr[2] = {1,2};  
    modify(arr);  
    printf("%d\n", arr[0]);  
}
```

Round 2

(Individual
Answering
/
Challenge)



```
int arr[3] = {5, 10, 15};  
int *p = arr;  
printf("%d\n", *p + *(p+2));
```

```
int x = 7;  
int *p = &x;  
*p += 3;  
printf("%d\n", x);
```

```
void foo(int arr[])  
{  
    arr[1] += 5;  
}
```

```
int main()  
{  
    int arr[3] = {2,4,6};  
    foo(arr);  
    printf("%d\n", arr[1]);  
}
```

**(BONUS
ROUND** / Curiosity / Open
Discussion)

Round 3

```
int *p;  
printf("%d\n", *p);
```

```
int a = 10;  
int *p = &a;  
int **q = &p;
```

```
**q = 20;
```

```
printf("%d %d %d\n", a, *p, **q);
```

```
void change(int *p) {  
    *p = 50;  
}
```

```
int main() {  
    int a = 10;  
    change(&a);  
    printf("%d", a);  
    return 0;  
}
```


Lec 2 plan [interactive live coding]

Garbage value: Its not a random value, it is someone else's old life. AirBNB vs Raddison: Every house has a name (Mannat)

Then print address: change format specifier (needed). Nasa once lost a satellite due to unit match. In C you can loose a program due to this. It doesnt stop you rather lets you crash gracefully.

Use a new pointer variable to print address (Show referencing dereferencing): Confusing pointer syntax: First impression is not the last impression

Need 100 vars, x1, x2, x3,... x100. Your compiler will nit stop you. It will just loose respect for you. You are thinking arrays, lets take a step back.

*(p+1): To discuss, operator precedence, pointer arithmetic, legal/illegal memory, if its unclaimed you will get it but is risky, like that guy in modinagar tied his animals to an empty plot.

An array allows you to do exactly this!! But legally

Traditional use of Array vs Pointer based which we won't use

Passing arrays to functions, how its different than variable passing

1. C doesn't protect you — it trusts you
2. Address $\neq$ Value
3. * lies on first impression
4. Arrays are **legal pointer arithmetic**
5. Passing arrays $\neq$ passing variables